

ranging from 402 to 405MHz, is available for use by implanted medical devices to provide high-speed communication between patients' IECs and medical practitioners. Choice of the frequency is made based on associated body losses and antenna efficiency. The 400MHz region offers optimisation in this respect with better ability to transmit radio signals in the human body.

However, IECs are susceptible to electromagnetic interference (EMI) and radio frequency interference (RFI) when exposed to electromagnetic fields or radiations from radio transmitters. Potential sources include high-voltage transformers, powerful electromagnets, radio transmitters, and various types of electronic equipment and devices, such as mobile phones and antitheft devices.

The interference risk depends on factors such as distance from the source, transmitted power, and the level of protection provided in the device itself. The malfunctions of IECs include over/under sensing, induction of asynchronous ventricular dysrhythmia, and electronic noise. As a safeguard, precautions are required to be taken by the patient, such as using cellular phone in the ipsilateral ear and maintaining distance from high-voltage transformers, machines, and gadgets/ devices emitting strong electromagnetic radiations.

## The systems available

Almost all the manufacturers are marketing proprietary IECs that are compatible with their own remote monitoring systems and protocols only. As a result, collection and management of data requires manufacturer-specific equipment. Although there are operational differences, the general flow of information is similar. Some of the systems in use are:

**Home monitoring.** Biotronik, a German firm, was the first to intro-

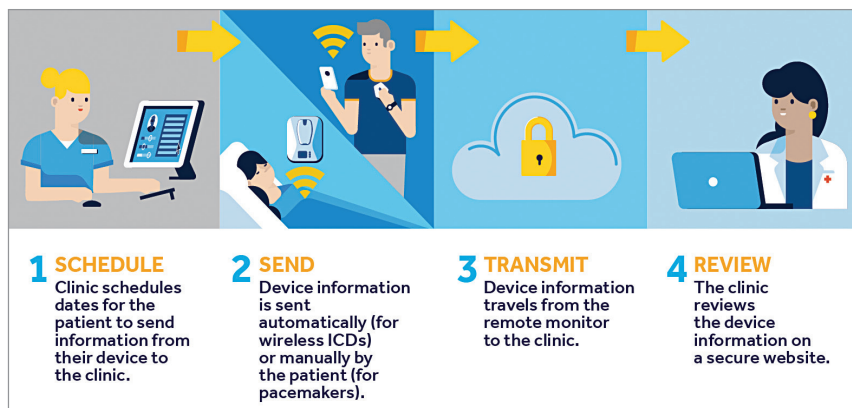


Fig. 6: Remote monitoring steps (Source: www.medtronic.com)

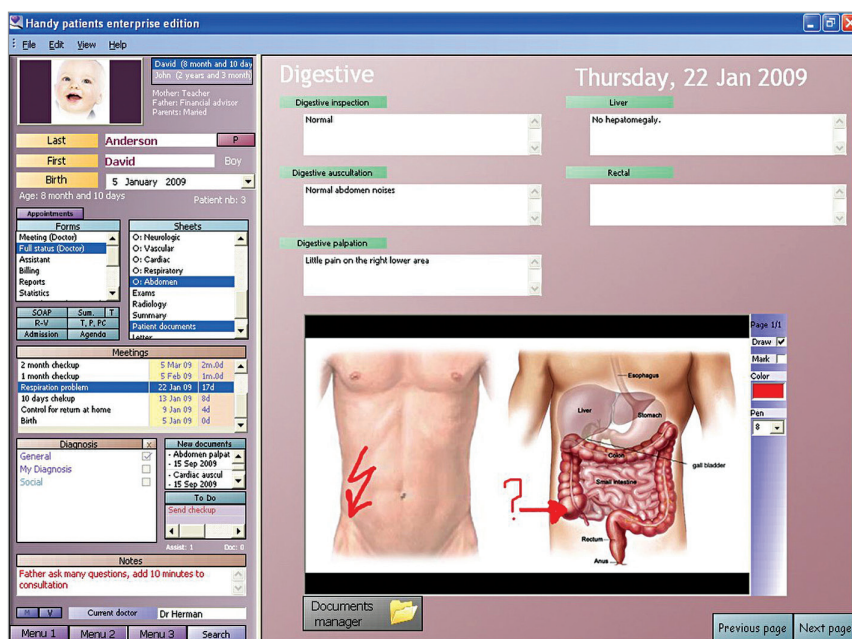


Fig. 7: Example of an electronic medical record (Source: www.medtronic.com)

duce their remote home monitoring system in 2001. The system transmits data on a daily basis at fixed time intervals and on the occurrence of a clinically relevant event. A cell-phone-like patient device sends encrypted data automatically via mobile phone to a central database in Europe. On reception, the data is processed and made available to the caregiver.

**Caretaker network.** The CareLink network of Medtronic, USA, is one of the leading remote monitoring systems, which can be used by a compatible Medtronic patient implanted heart device. The system

is customisable and flexible, and has a monitor that interfaces with the device and a telecommunications link to transmit data to the healthcare team.

CareLink also enables display of diabetes information, such as continuous glucose monitoring and insulin pump data. The network enables the healthcare team to access the information on a secure website or through a mobile app. Colour-coded notifications are made available to the patients via options that include website, voice message, pager, text message, or email message.